

Alessandro Metelli

Software Engineer — Backend · Developer Infrastructure · AI Tooling & Agent Workflows

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SUMMARY

Software Engineer with 5 years of experience building backend systems, developer infrastructure, blockchain tooling, DeFi protocols, and AI tooling for agentic workflows. Strong in Rust, TypeScript, Python, Go, distributed systems, and developer productivity tooling. Recently built AgentReceipt, Skills Doctor, and RitualAI: local-first agent/developer tools for AI coding observability, skill quality auditing, and reusable workflow generation. Seeking fully remote backend, platform, developer tools, or AI tooling roles.

EXPERIENCE

Nethermind — Software Engineer

Apr 2024 – Jun 2026 · Remote

- Built backend systems and developer tooling in Rust, TypeScript, Python, Cairo, and Solidity for infrastructure, verification, automation, and developer workflows.
- Developed AggSandbox, a Rust CLI for local multichain infrastructure, transaction execution, bridge testing, and debugging across Starknet, AggLayer, and EVM environments.
- Maintained Voyager Verifier, improving Starknet contract verification through compiler compatibility work, metadata validation, clearer diagnostics, and workflow reliability improvements.
- Contributed sncast verify functionality to Starknet Foundry, expanding contract verification support for Starknet developers.
- Built integrations across Starknet, LayerZero, Hyperlane, AggLayer, and EVM ecosystems for cross-chain messaging, local testing, and infrastructure validation.
- Developed ZK infrastructure components, including Groth16 verification and Ethereum-to-Starknet data proving systems.
- Built AI-assisted engineering workflows and tools focused on automation, validation, observability, traceability, controlled execution, and developer productivity.

ScopePilot — Founder & Lead Developer

2026 – Present · Remote

- Built an agency proposal scope and pricing configurator with proposal workflows, pricing inputs, client delivery, approvals, revisions, and reporting.
- Implemented a TypeScript/Bun product with Astro, React, Hono, PostgreSQL, Drizzle, Stripe billing, Cloudflare deployment, and automated quality gates.

SELECTED PROJECTS

AgentReceipt: Local-first Go CLI sidecar for AI coding sessions: watches Codex activity, records tool calls/commands/edits/tokens/warnings, and produces signed, hash-chained review receipts. **Tech:** Go, CLI, Git, Ed25519, observability

Skills Doctor: TypeScript/Bun CLI that audits Claude/Codex Agent Skills for spec compliance, quality scoring, referenced-file integrity, JSON reports, and repair handoff. **Tech:** TypeScript, Bun, Vitest

RitualAI: TypeScript CLI that scans local Claude/Codex prompt history, clusters recurring workflows, and turns approved patterns into reusable SKILL.md files with validation and local-first privacy. **Tech:** TypeScript, Bun, Node.js

ScopePilot: Agency proposal scope and pricing configurator with private proposal delivery, billing, reporting, and launch-readiness validation. **Tech:** TypeScript, Bun, Astro, React, Hono, PostgreSQL, Cloudflare

OSS CONTRIBUTIONS

- **Voyager Verifier / Starknet Foundry / Dojo Engine / Apibara DNA:** 47 merged Starknet OSS PRs across Rust verification tooling, Cairo ERC/DeFi primitives, Dojo support, indexing/devnet integration, docs, releases, and PostgreSQL sink infrastructure.
- **Apache DataFusion:** Merged SQL planner diagnostics for mistaken NULL equality predicates and struct-returning aggregate UDF docs/examples/tests for window metadata projection.
- Additional OSS PRs in Supabase CLI, PostHog, OpenTelemetry Collector, and Apache Airflow covering database CLI workflows, data-warehouse schema reliability, HTTP compression hardening, and orchestration metadata/OpenLineage integration.

TECHNICAL SKILLS

Languages: Rust, TypeScript, JavaScript, Python, Go, Solidity, Cairo, Noir

Backend: Node.js, REST APIs, PostgreSQL, Redis, event-driven systems, distributed systems

Infrastructure: Docker, Linux, GitHub Actions, CI/CD, AWS **Frontend:** React, TypeScript

AI Tooling: Agent observability, signed receipts, prompt-history mining, skill generation/auditing, tool orchestration, validation, human-in-the-loop systems

Blockchain: Starknet, Ethereum, Aztec, LayerZero, Hyperlane, AggLayer, smart contracts, DeFi, ZK systems

EDUCATION

BSc Data Science and Computing — Birkbeck, University of London